

Practice Quiz: Agents (Implementation)

Name: __ Date: __

Part A: Multiple Choice

1. The `ActiveInferenceAgent` class stores beliefs in: A) `self.A` B) `self.qs` — the posterior distribution over hidden states C) `self.C` D) `self.gamma`
2. The `infer_states` method implements: A) Policy evaluation B) Bayesian updating: $\text{posterior} = \text{normalize}(\text{likelihood} \times \text{prior})$ C) Action selection D) Environment simulation
3. The `_compute_efe` method calculates: A) The observation likelihood B) Expected Free Energy by summing pragmatic and epistemic terms over predicted futures C) The transition probability D) The number of policies
4. The `SimpleEnvironment` generates observations by: A) Always returning the same value B) Sampling from $P(o | s)$ using the true `A` matrix and current state C) Reading from a file D) Copying the agent's beliefs
5. The simulation loop connects agent and environment through: A) Shared memory B) `Observe → Infer → Select → Act → Repeat` C) A single function call D) Random steps
6. Setting `gamma = 0` would cause the agent to: A) Always select the best policy B) Select policies nearly uniformly (random behavior) regardless of EFE C) Crash the program D) Move faster
7. The agent logs `belief_history`, `action_history`, and `obs_history` for: A) Entertainment B) Analysis and debugging — understanding the agent's decision-making over time C) Sending to another agent D) Reducing computation

Part B: Short Answer

1. Write the key lines of the `infer_states` method and explain what each line does.

2. Explain what happens in the simulation loop when the agent's model (A matrix) is slightly wrong — different from the environment's true A matrix. How does this affect behavior?
3. Design a test to verify that the agent's policy selection is working correctly. What should happen when C strongly favors one observation? When C is uniform?